

ANSWER KEY WITH EXPLANATIONS

MEASUREMENT AND UNIT CONVERSIONS

Part A: Fill in the Blanks

1. Unit
2. Physical
3. Fundamental
4. Derived
5. Metre (m)
6. Kilogram (kg)
7. Second (s)
8. Cubit
9. Handspan
10. Systeme International d'Unites (SI)

Part B: Choose the Correct Answer

1. d) Volume
2. c) CGS
3. b) Thermometer
4. c) Foot
5. b) 1000 g
6. d) kilo
7. c) Area
8. b) Foot–Pound–Second
9. c) Volume
10. b) Magnitude

Part C: Conversion Practice

1. Convert 7 km into metres.

Given: 7 km

$$1 \text{ km} = 1000 \text{ m}$$

$$7 \times 1000 = 7000 \text{ m}$$

Answer: 7000 m

2. Convert 4500 m into kilometres.

Given: 4500 m

$$1000 \text{ m} = 1 \text{ km}$$

$$4500 \div 1000 = 4.5 \text{ km}$$

Answer: 4.5 km

3. Convert 9 m into centimetres.

Given: 9 m

$$1 \text{ m} = 100 \text{ cm}$$

$$9 \times 100 = 900 \text{ cm}$$

Answer: 900 cm

4. Convert 750 cm into metres.

Given: 750 cm

$$100 \text{ cm} = 1 \text{ m}$$

$$750 \div 100 = 7.5 \text{ m}$$

Answer: 7.5 m

5. Convert 12 m into millimetres.

Given: 12 m

$$1 \text{ m} = 1000 \text{ mm}$$

$$12 \times 1000 = 12000 \text{ mm}$$

Answer: 12000 mm

6. Convert 8500 mm into metres.

Given: 8500 mm

$$1000 \text{ mm} = 1 \text{ m}$$

$$8500 \div 1000 = 8.5 \text{ m}$$

Answer: 8.5 m

7. Convert 6 km 250 m into metres.

$$6 \text{ km} = 6 \times 1000 = 6000 \text{ m}$$

$$6000 \text{ m} + 250 \text{ m} = 6250 \text{ m}$$

Answer: 6250 m

8. Convert 345 cm into millimetres.

$$1 \text{ cm} = 10 \text{ mm}$$

$$345 \times 10 = 3450 \text{ mm}$$

Answer: 3450 mm

9. Convert 8 kg into grams.

$$1 \text{ kg} = 1000 \text{ g}$$

$$8 \times 1000 = 8000 \text{ g}$$

Answer: 8000 g

10. Convert 7500 g into kilograms.

$$1000 \text{ g} = 1 \text{ kg}$$

$$7500 \div 1000 = 7.5 \text{ kg}$$

Answer: 7.5 kg

11. Convert 5 kg 250 g into grams.

$$5 \text{ kg} = 5000 \text{ g}$$

$$5000 \text{ g} + 250 \text{ g} = 5250 \text{ g}$$

Answer: 5250 g

12. Convert 4500 mg into grams.

$$1000 \text{ mg} = 1 \text{ g}$$

$$4500 \div 1000 = 4.5 \text{ g}$$

Answer: 4.5 g

13. Convert 3 quintals into kilograms.

$$1 \text{ quintal} = 100 \text{ kg}$$

$$3 \times 100 = 300 \text{ kg}$$

Answer: 300 kg

14. Convert 2 tons into kilograms.

$$1 \text{ ton} = 1000 \text{ kg}$$

$$2 \times 1000 = 2000 \text{ kg}$$

Answer: 2000 kg

15. Convert 3 hours into minutes.

$$1 \text{ hour} = 60 \text{ minutes}$$

$$3 \times 60 = 180 \text{ minutes}$$

Answer: 180 min

16. Convert 180 minutes into hours.

$$60 \text{ min} = 1 \text{ h}$$

$$180 \div 60 = 3 \text{ h}$$

Answer: 3 h

17. Convert 2 days into hours.

$$1 \text{ day} = 24 \text{ h}$$

$$2 \times 24 = 48 \text{ h}$$

Answer: 48 h

18. Convert 4 hours 20 minutes into minutes.

$$4 \text{ h} = 4 \times 60 = 240 \text{ min}$$

$$240 \text{ min} + 20 \text{ min} = 260 \text{ min}$$

Answer: 260 min

19. Convert 5400 seconds into hours.

$$1 \text{ h} = 3600 \text{ s}$$

$$5400 \div 3600 = 1.5 \text{ h}$$

Answer: 1.5 h or 1 h 30 min

20. Convert 2 years into days.

$$1 \text{ year} = 365 \text{ days}$$

$$2 \times 365 = 730 \text{ days}$$

Answer: 730 days

Part D: Word Problems

1. A cloth merchant has 12 m of cloth. He sells 450 cm. How much cloth is left?

$$450 \text{ cm} = 450 \div 100 = 4.5 \text{ m}$$

$$\begin{aligned} \text{Remaining cloth} &= 12 \text{ m} - 4.5 \text{ m} \\ &= 7.5 \text{ m} \end{aligned}$$

Answer: 7.5 m

2. A milkman delivers 25 litres of milk every day. How many litres does he deliver in 7 days?

$$25 \times 7 = 175$$

Answer: 175 L

3. A scrap dealer collected 45 kg of newspapers on Monday and 38 kg on Tuesday. What is the total mass collected?

$$45 + 38 = 83$$

Answer: 83 kg

4. A student walks 2 km to school and 2 km back home. How many metres does the student walk in a day?

$$2 \text{ km} + 2 \text{ km} = 4 \text{ km}$$

$$4 \times 1000 = 4000 \text{ m}$$

Answer: 4000 m

5. A train travels 150 km in the morning and 175 km in the evening. Find the total distance travelled in metres.

$$150 + 175 = 325 \text{ km}$$

$$325 \times 1000 = 325000 \text{ m}$$

Answer: 325000 m

6. A doctor checks the temperature of 15 patients in one hour. How many patients can be checked in 4 hours?

$$15 \times 4 = 60$$

Answer: 60 patients

7. A farmer harvested 3 quintals of wheat from one field and 2 quintals from another field. Find the total mass in kilograms.

$$3 + 2 = 5 \text{ quintals}$$

$$5 \times 100 = 500 \text{ kg}$$

Answer: 500 kg

8. A water tank contains 500 litres of water. If 125 litres are used, how much water remains?

$$500 - 125 = 375$$

Answer: 375 L

9.

$$1 \text{ km} = 1000 \text{ m}$$

To convert metres into kilometres, divide by 1000.

$$20.6 \div 1000 = 0.0206$$

Answer: 0.0206 km

10

$$1 \text{ m} = 1000 \text{ mm}$$

To convert metres into millimetres, multiply by

1000.

$$\begin{array}{r} \frac{3}{20} \times 1000 \\ = \frac{3000}{20} \\ = 150 \end{array}$$

Answer: 150 mm

11

$$1 \text{ km} = 1000 \text{ m}$$

$$0.005 \times 1000 = 5 \text{ m}$$

$$1 \text{ m} = 10 \text{ dm}$$

$$5 \times 10 = 50 \text{ dm}$$

Answer: 50 dm

12

$$1 \text{ dm} = 100 \text{ mm}$$

To convert millimetres into decimetres, divide by 100.

$$3000 \div 100 = 30$$

Answer: 30 dm

MINDFUL EATING

A. Multiple Choice Questions

1. c) Carbohydrates
2. c) Vitamin C
3. c) Body-building foods
4. b) Digestion
5. c) All nutrients, water and roughage in proper proportion

B. Fill in the Blanks

1. food
2. Proteins
3. protective
4. nutrients
5. nutri

1. Aarav often eats chips, noodles and soft drinks.

a) Which type of nutrients may be missing from his diet?

Vitamins, minerals and proteins may be missing from his diet.

b) Explain any two health problems that may occur due to this diet.

- He may become weak and tired due to lack of nutrients.
- He may fall sick frequently because his immunity becomes weak.

c) Suggest two foods he should include in his meals.

- Fresh fruits
- Green leafy vegetables

(Other acceptable answers: milk, eggs, pulses, nuts)

2. Meera has difficulty seeing clearly in dim light.

a) Which vitamin is Meera lacking?

Vitamin A

b) Name the deficiency disease caused by its deficiency.

Night Blindness

c) Mention two functions of this vitamin.

- Helps maintain healthy eyesight.
- Keeps the skin healthy.

d) Name any two other food sources rich in this vitamin.

- Papaya
- Sweet potato

(Other acceptable answers: pumpkin, milk, egg yolk)

3. Complete the Nutrition Detective Challenge.

Symptom	Deficiency Disease	Nutrient Needed
Bleeding gums	Scurvy	Vitamin C
Weak bones	Rickets	Vitamin D
Enlarged neck	Goitre	Iodine
Poor vision at night	Night Blindness	Vitamin A

Why does a balanced diet help prevent these diseases?

A balanced diet provides all essential nutrients in the correct amounts. It helps the body grow

properly, remain healthy and prevents deficiency diseases caused by lack of vitamins and minerals.

These diseases can be prevented by eating a balanced diet containing all essential vitamins and minerals.

4. Imagine you are a doctor.

a) Identify the deficient vitamin.

Vitamin D

b) Name the disease caused by its deficiency.

Rickets

c) Explain how sunlight helps the body.

Sunlight helps the skin produce Vitamin D, which is needed for strong bones and teeth.

d) Suggest three foods that can help the patient recover.

- Milk
- Egg yolk
- Fish

(Other acceptable answers: butter, cheese, fortified cereals)

6. Match the Disease with the Clue and Explain

Clue	Disease
Bones become soft and bent	Rickets
Gums bleed easily	Scurvy
Neck becomes swollen	Goitre
Difficulty seeing in dim light	Night Blindness

Nutrient deficiency responsible for each disease

- **Rickets** → Vitamin D deficiency
- **Scurvy** → Vitamin C deficiency
- **Goitre** → Iodine deficiency
- **Night Blindness** → Vitamin A deficiency

EXPLORING MAGNETS

A. Multiple Choice Questions

- 1. b) Magnetite
- 2. b) Magnetism
- 3. d) Plastic
- 4. c) Poles
- 5. b) North–South direction

B. Fill in the Blanks

- 1. Magnetite
- 2. Magnetic
- 3. North, South
- 4. Repel
- 5. Magnetism

C. Passage-Based Questions

- 1. Attractive property of magnets.
- 2. At the poles (ends) of the magnet.
- 3. North–South direction.
- 4. Unlike poles attract each other.

5. Repulsion is called the sure test of magnetism because only a magnet can repel another magnet. Attraction can occur between a magnet and a magnetic material, but repulsion occurs only between two magnets.

D. Table-Based Questions

- 1. Iron and Nickel.
- 2. Wood and Plastic.
- 3. Iron.
- 4. A magnet attracts iron nails because they are magnetic, while wood shavings are non-magnetic and are not attracted.

5. Magnetic materials are attracted by magnets, whereas non-magnetic materials are not attracted by magnets.

E. Answer the Following Questions

1. Differentiate between magnetic and non-magnetic materials.

Magnetic Materials	Non-Magnetic Materials
Attracted by magnets	Not attracted by magnets
Examples: Iron, Nickel, Cobalt	Examples: Wood, Plastic, Copper

2. Explain any three properties of magnets.

- i. **Attractive Property:** Magnets attract magnetic materials such as iron, nickel and cobalt.
- ii. **Directive Property:** A freely suspended magnet always points in the North–South direction.
- iii. **Attraction and Repulsion:** Like poles repel each other, while unlike poles attract each other.

3. What is a magnetic compass? Mention its uses.

A magnetic compass is a device containing a freely rotating magnetic needle.

Uses:

- Finding directions.
- Navigation by sailors and explorers.
- Trekking and adventure activities.

4. Why are the poles of a magnet important?

- Poles are the strongest parts of a magnet.
- Most magnetic attraction occurs at the poles.

- Attraction and repulsion take place through the poles.
- They help determine magnetic directions.

F. Answer the Following Questions

1. Scrap Dealer Activity

- a) Iron nails and steel pins.
- b) Plastic beads and wooden pieces.
- c) The magnet attracts iron nails and steel pins, separating them from non-magnetic materials.
- d) It is quick, easy and saves time and effort.

2. Magnet Detective Table

Observation	Property of Magnet
Iron filings stick mostly at the ends	Poles of a Magnet
A freely suspended magnet points north-south	Directive Property
N pole attracts S pole	Attraction
N pole repels N pole	Repulsion

Explanation:

These properties help magnets in compasses, separating materials, refrigerator doors and many electrical devices.

3. Explorer in a Forest

- a) The compass helps find directions.
- b) The magnetic needle aligns itself in the North–South direction due to Earth's magnetic field.
- c) Earth behaves like a giant magnet, which causes the compass needle to point north-south.
- d) Before GPS, sailors, travellers and explorers used compasses for navigation and finding directions.

4. Breaking a Magnet

- a) Each piece becomes a smaller magnet with both North and South poles.
- b) No. A single North pole cannot exist alone because magnetic poles always occur in pairs.
- c) Magnetic poles always exist together as North and South poles.
- d) Breaking a magnet produces smaller magnets, each having both poles.

5. Preserving Magnets Challenge

- a) The magnet may lose some of its magnetic strength.
- b) Demagnetisation is the process by which a magnet loses its magnetic properties.
- c) Precautions:
 1. Do not drop magnets.
 2. Do not heat magnets.
 3. Store magnets with soft iron keepers (creepers).
- d) Careful storage prevents loss of magnetism and increases the life of the magnet.

AMAZING WORLD OF SCIENCE & DIVERSITY IN THE LIVING ORGANISMS

Section A: Multiple Choice Questions

1. b) Observation
2. b) Hypothesis
3. c) Plant
4. c) Respond to stimuli
5. a) Virus
6. c) Solid
7. c) Evaporation
8. c) Carbohydrate
9. c) Sand
10. b) Frog

Section B: Fill in the Blanks (5 × 1 = 5 Marks)

11. Habitat
12. Venation
13. Monocotyledons (Monocots)
14. Taproot
15. Locomotion

Section C: Very Short Answer Questions (5 × 2 = 10 Marks)

16. Define biodiversity.

Biodiversity is the variety of different living organisms found on Earth.

17. What is a hypothesis?

A hypothesis is an educated guess that can be tested through experiments.

18. Name any two aquatic animals.

Fish, Whale / Dolphin / Octopus (any two)

19. What is condensation?

Condensation is the process by which water vapour changes into liquid water.

20. Give two examples of shrubs.

Rose and Hibiscus (Lemon is also correct)

Section D: Short Answer Questions

21. Differentiate between taproot and fibrous root with examples.

Taproot	Fibrous Root
Has one main root with smaller side roots.	Has many thin roots of similar size.
Grows deep into the soil.	Spreads near the surface of the soil.
Example: Carrot, Mango	Example: Grass, Wheat

22. Explain any three characteristics of living things.

- Living things grow and develop.
- Living things reproduce their own kind.
- Living things respond to stimuli in their surroundings.
- Living things need food and energy. (Any three)

23. Why do scientists classify living organisms? Give any three reasons.

- Makes organisms easier to study.
- Helps understand relationships among organisms.
- Avoids confusion caused by different local names.

Section E: Long Answer Questions (2 × 3 = 6 Marks)

24. Explain any three adaptations of a cactus that help it survive in the desert.

- Leaves are modified into spines to reduce water loss.

- Thick, fleshy stem stores water.
- Deep roots absorb water from underground.
- Waxy coating reduces evaporation. (*Any three*)

25. Camels are called the “Ships of the Desert.” Explain any three adaptations that help them survive in desert conditions.

- Hump stores fat for energy.
- Long eyelashes protect eyes from sand.
- Closable nostrils prevent sand from entering.
- Padded feet help walk on sand.
- Long legs keep the body away from hot sand.
- They conserve water by sweating very little. (*Any three*)